

Input image

**HGNetv2-B4**
Hybrid encoder

**Decoder layers 1–5**

Feature maps

Side probabilities

Deformable cross-attention + gateway

**Exterior mean**

$$F' = F - 0.5m$$

Content and Δ

Decoder replay (training only)

**BCSA**
Expected region state

$q$

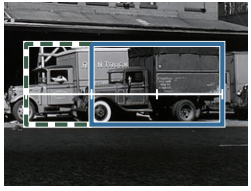
State correction

$q^+$

**FFN + heads**

## (a) From candidate rectangles to a query update

**Candidate rectangles**



-- **A** wide — **B** tight

**Interval states**

| Cell write | 4 | 2 | 3 | Box prob. |
|---|---|---|---|---|
| **A** [1, 3] | 4 → | 4 → | 5 | × ¼ |
| **B** [2, 3] | – | 2 → | 4 | × ¾ |

$h_j = \frac{1}{2} h_{j-1} + w_j$ reset at each left edge

**Boundary-weighted read**

$\bar{H}_q = \frac{1}{4}(5) + \frac{3}{4}(4) = 4.25$

$q$ Observed query

$\Sigma \rightarrow \Phi(q, \bar{H}_q) \rightarrow (+) \rightarrow q^+$

Project the state and add it to the query.

## (b) Using the same shifted feature for content and step size

**Encoder grid**

**Encoder memory** $F$

Reuse the query $q$ and box probabilities $\pi_q$

Shaded: box exterior

$F' = F - 0.5m$

$m$: exterior mean

Content $v', u'$ → $w' = \Delta' v' u'$ write

Step size $\Delta'$ → $a' = e^{-\Delta' e^{\theta}}$ retain

**BCSA** → FFN, heads → $0.1\,\mathcal{L}_{\text{aux}}$

Training targets